# DEVELOPMENT OF AN AUTOMATIC SPEECH-TO-SPEECH SYNTHESIS ARCHITECTURE FOR ENGLISH-TO-YORÙBÁ LANGUAGE USING NEURAL MACHINE TRANSLATION APPROACH

*Saturday Presentation (ASPMIR)*

Presented By

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# OUTLINE

# 1. ABSTRACT

- Different researchers have developed various tools and applications to ensure seamless communication between human beings for day-to-day transactions and interactions.
- Speech-to-Speech synthesis entails translating the speech of one language to the speech of another language.
- This proposal aims to launch the development of speech-to-speech resources using the neural machine translation approach.
- The focus of the work will be translational use case between a medical practitioner and patient as well as public health practitioner and public.
- An extension to Translatotron 2 architecture will be developed in this proposed work to translate from English to Yoruba languages.
- To develop the proposed model, Yorùbá-English general domain and medical domain speech corporal will be curated, and two models will be trained and developed.
- These are the base model and the transfer learning model using an already pre-trained English language model to another language.
- What follows is the training of the two models using the general domain speech corporal of Yorùbá-English curated.
- Second is the focus of the work, where the medical domain data will be utilised to train the two models.
- Thereafter comparison of the models developed under different experiments will be carried out.
- Finally, a prototype Softbot shall be developed and evaluated.

# 2. INTRODUCTION

## 2.1 Background

- English language is the third most widely spoken native language in the world behind Mandarin Chinese and Spanish, as well as the most spoken language in the world spoken by nearly 1.5 billion speakers, which is slightly higher than 1.1 billion Mandarin Chinese speakers (Silaban et al., 2023).

- Yoruba is a language that shares membership with the Niger-Congo language family (Abiola et al., 2014) and one of the three official languages in Nigeria, spoken by a population of about 50 million across South Western Nigeria, coupled with the USA, UK, Brazil, Benin and Togo (Akintola & Ibiyemi, 2017).

- The translation of speech from one language to another is one of the research areas in speech synthesis that has gained recognition in recent years (Chang et al., 2023), and different approaches have been utilised to achieve this task.

- The approaches are illustrated in Figure 1, while Figures 2 to 5 show the models available for each component of the cascaded system.

- **United Nations SDGs**
  - ✓ **Goal 3 – Ensuring healthy lives and well-being**
  - ✓ **Goal 10 – reducing inequality within and among the community.**
- **Industrial Revolution 4(IR4)**

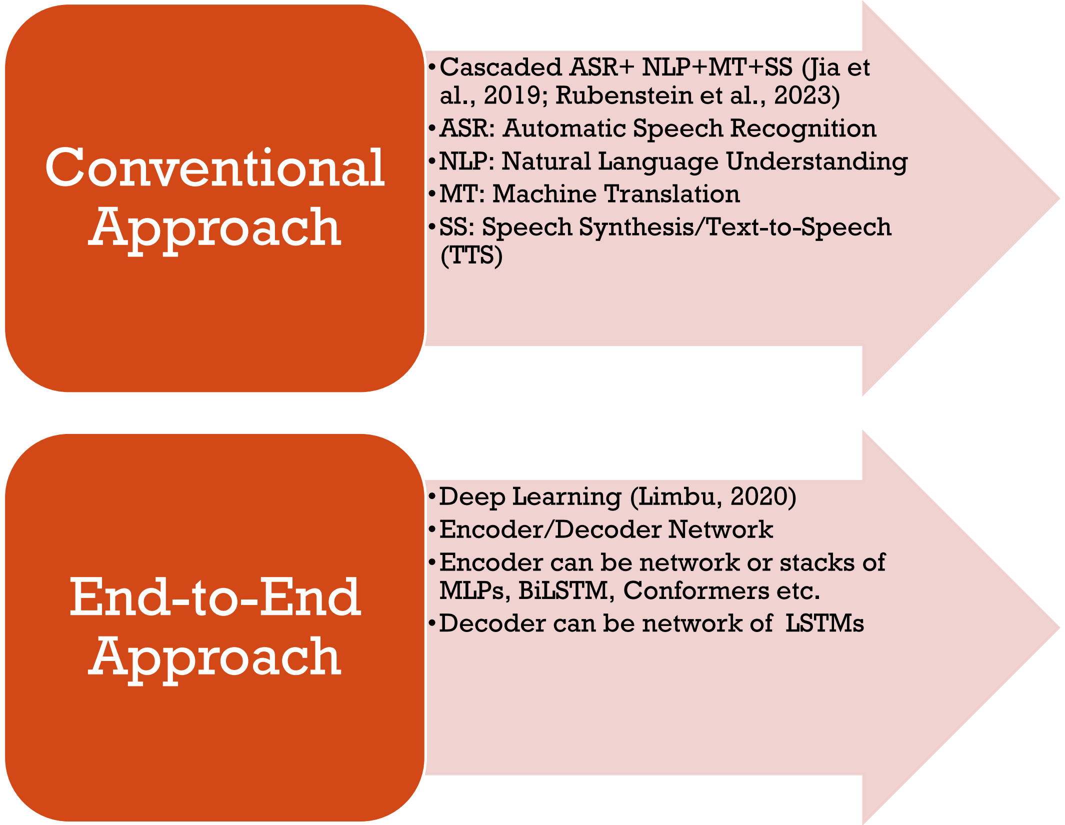



**Figure 1:** Approaches to Modelling Speech-to-Speech Synthesis

# 2. INTRODUCTION

## 2.1 Background (Cont.)

- **Well-known S2ST models developed**

- Seamless M4T (Barrault et al., 2023)
- Whisper (Radford et al., 2023a)
- AudioPalm (Rubenstein et, al., 2023)
- Translatotron (Jia, Weiss, et al., 2019)
- mSLAM-CTC
- Maestro

**Figure 2:** Well-known S2ST Models

- **Well-known MT models developed**

- Meta (R. Li et al., 2020)
- T5 (Text-to-Text Transfer Transformer) (Xue et al., 2020)
- MarianNMT (Rapp, 2021)
- RoBERTa (Xue et al., 2020)
- mBERT (Xue et al., 2020)
- XLM (Xue et al., 2020)
- OpenNMT
- LibreTranslate

**Figure 3:** Well-known MT Models

- **Well-known ASR models developed**

- SeamlessM4T (Barrault et al., 2023)
- Jasper (Li et al., 2019a)
- Whisper (Radford et al., 2023a)
- Translatotron (Jia, Weiss, et al., 2019)
- Translatotron2 (Jia et al., 2022)
- Bidirectional Encoder Representative of a Transformer (BERT)

**Figure 4:** Well-known ASR Models

- **Well-known TTS models developed**

- WaveNet (Oord et, al., 2018)
- Tacotron 2 (de Martos, 2022)
- **MelGAN (Kumar et, al., 2019)**
- FFNET (Jin et. al., 2018)
- **Parallel WaveGANs (Yamamoto et, al., 2020)**
- LPC-Net (Valin & Skoglund, 2019)

**Figure 5:** Well-known TTS Models

# 2. INTRODUCTION

## 2.2 Problem Statements

- The number of available speech resources or tools that are not sufficient as well as Yorùbá speech corpus. (Bansal, 2019; Zhou et al., 2013, Fagbolu et al., 2015; Meyer et al., 2022) .
- The output audio samples of the target language are not real representation of the input audio samples (Limbu, 2020).
- Medical speech corpora are not available for training (Cherednichenko, 2020; Grabar & Cardon, 2019.; Qian et al., 2023).
- Language barriers (Hudelson & Chappuis, 2024; Javaid et al., 2023; Panayiotou et al., 2020; Tu et al., 2024)
- The uncertain alignment between the input and output spectrograms is challenging (Jia, Weiss, et al., 2019).
- Delays and errors are associated with the cascaded system.

## 2.3 Justification for the Study

- Speech-to-speech translation can minimize the language barriers experienced by individuals in their conversations with colleagues who speak other languages.
- Direct speech-to-speech translation to be considered in this work has the benefit of preserving the prosody, emotion, accent, and voice of the speaker.
- More speech data for English-Yoruba language pairs will be publicly available.

# 2. INTRODUCTION

## 2.4 Aim and Objectives

- The aim of this research is to develop an automatic Speech-to-Speech Synthesis Model for English-to-Yorùbá Language based on Neural Machine Translation Approach.

- The objectives to realise the aim are:

I. create general and public health speech corpus for English-to-Yorùba language pair with focus on the accent of speakers of the two languages.

II. preprocessing of the data collected in steps i, and ii

III. design and implement an improved architecture for speech-to-speech translation (S2ST) of English-to-Yoruba language pair.

IV. develop a prototype Softbot based on the S2ST architecture in (iii) for malaria public health education use case.

V. carry out performance evaluation of the S2ST architecture and the prototype Softbot.

# 2. INTRODUCTION

## 2.5 Scope of the Study

- Improved speech-to-speech synthesis model from English-to-Yorùbá language translation will be developed.
- The parallel speech corpora, will be curated and to be made available for research community.
- The corpora will include general and public health domains.
- A prototype Softbot based on the S2ST model will be developed

## 2.6 Expected Contribution to Knowledge

1. **General and public health corpus for English-to-Yoruba direct S2ST corpus will be created and open-sourced for the language technology research community.**
2. **Improved data preprocessing algorithms for audio corpus**
3. **An improved direct S2ST architecture for English-to-Yoruba that can be finetuned for other low-resourced African language pairs.**
4. **A crowd-sourcing web application and a prototype S2ST Softbot that could be used for translation within the public health education domain and adapted for other public address translation use cases.**

# 3. LITERATURE REVIEW

## 3.1 Generic Architecture of Direct S2ST

❑ **The architecture of a direct S2ST is illustrated in Figure 6 (Jia et al., 2019; Jia et al., 2022)**

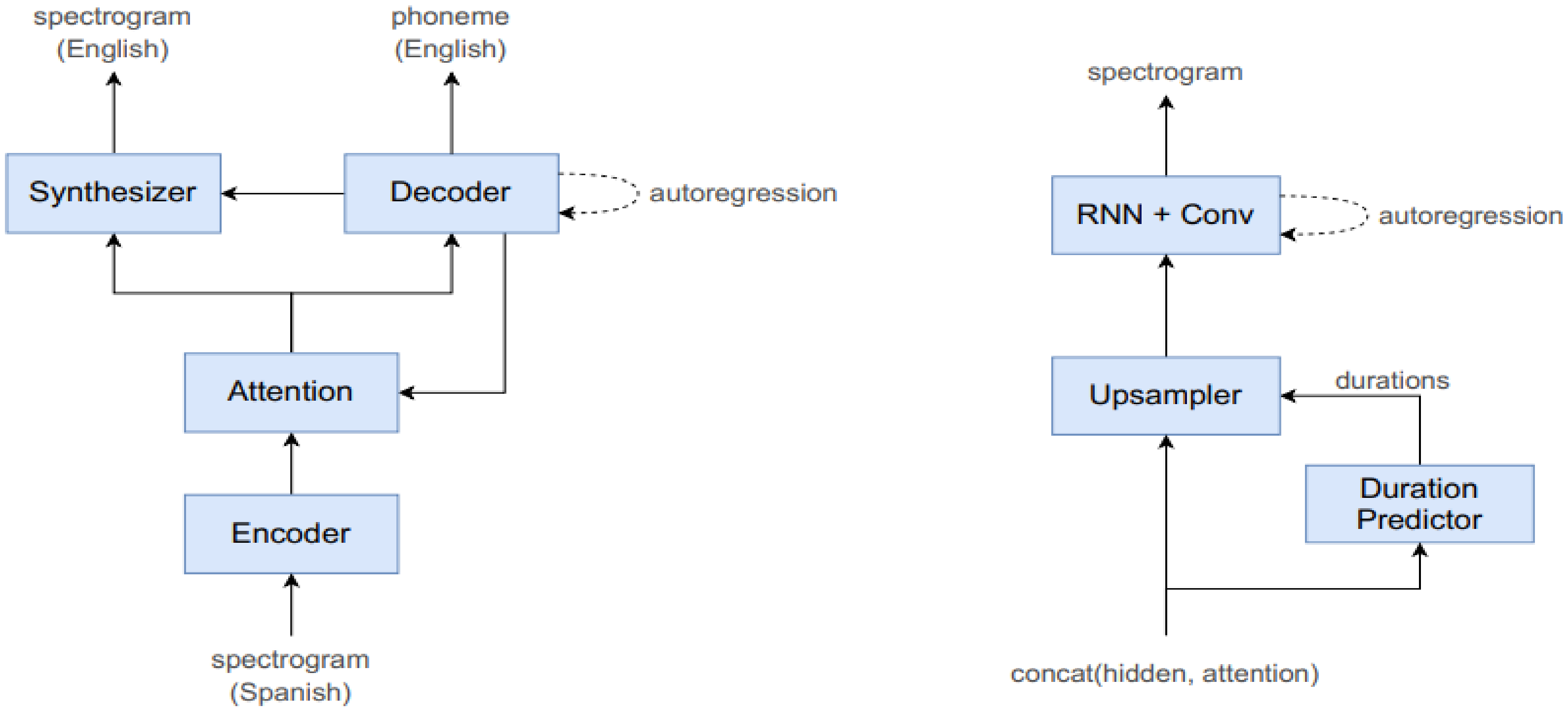



**Figure 6:** Generic Direct Speech-to-Speech Architecture (Translatotron 2) (Jia et al., 2022)

## 3.2 Transformer Architecture of Direct S2ST

- ❑ **Model will also be developed for transformer architecture is illustrated in Figure 7 (Vaswani et al., 2017)**

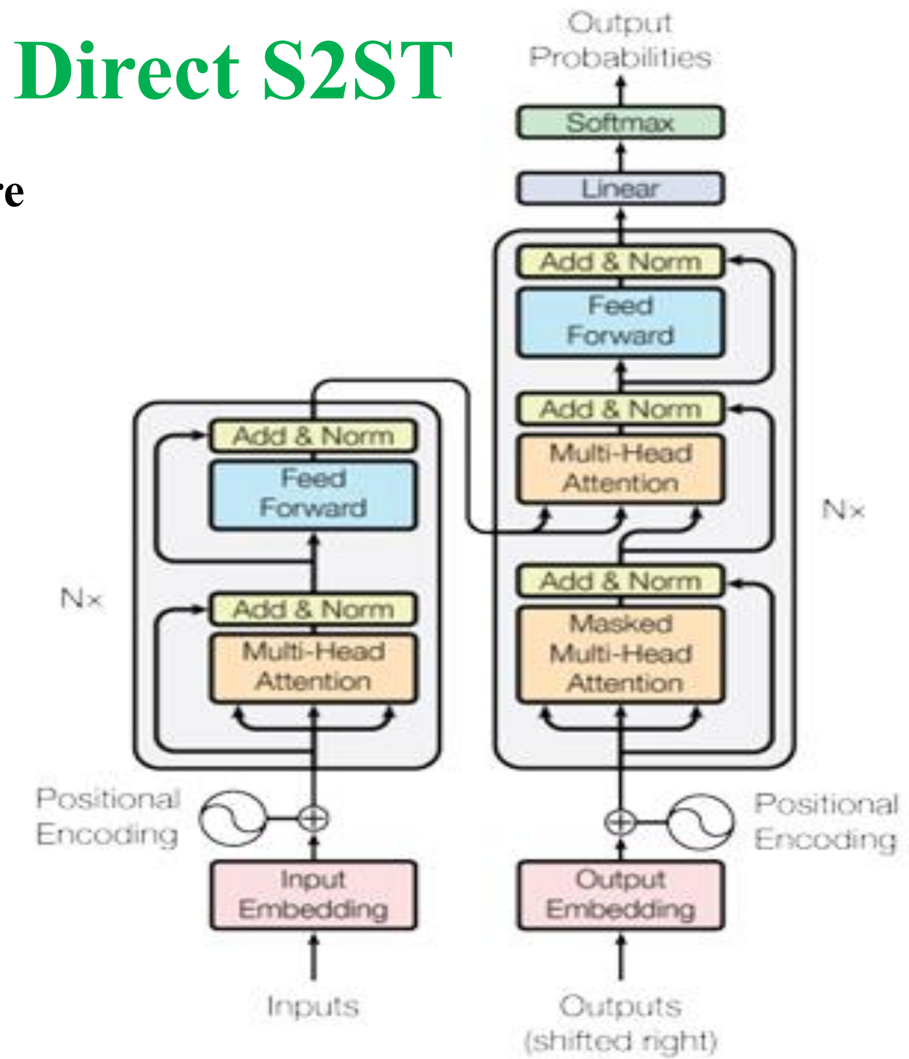



**Figure 7:** Transformer Architecture (**Vaswani** et al., 2017)

## 3.3 Summary of Literature Review

**Table 1:** Review of Related Works

| Ref | METHODS | DATASETS/ LANGUAGE | RESULTS | FEAATURE EXTRACTION METHODS | ADVANTAGES / LIMITATIONS |
|---|---|---|---|---|---|
| (Chang et al., 2023) | Sqeuence-to-Sequencye modelling using Convolution Sub-Sampling<br><br>ESPnet Toolkit for implementation | Dataset:<br>MuST-C dataset:<br>Average of 430 hours of audio recordings of TED talks.<br>Languages:<br>English-German, Spanish, French | BLEU scores using FBaank:<br>En-De: 26.7<br>En-Es: 31.2<br>En-Fr: 37.1<br><br>BLEU scores using Discrete tokens:<br>En-De: 28.6<br>En-Es: 33.0<br>En-Fr: 38.7 | (1) Use of discretisation for speech extracted features.<br>(2) Use of De-duplication and sub-word modelling to further reduce the size of the speech features. | Advantages<br>1. Performance enhancement<br>2. Reduction in training time due to discrete token.<br>Limitations<br>1. The work did not consider the comparison of other discretisation approach |

# 3. LITERATURE REVIEW (CONT.)

## 3.3 Summary of Literature Review

**Table 1:** Review of Related Works (Cont.)

| Ref | METHODS | DATASETS/LANGUAGE | RESULTS | FEAATURE EXTRACTION METHODS | ADVANTAGES / LIMITATIONS |
| --- | --- | --- | --- | --- | --- |
| (Jia et al., 2019) | **Attention Sequence-to-Sequence approach** with: Speaker Encoder, and Neural Vocoder<br><br>**Translatotron** | **Datasets:**<br><br>**Conversational Spanish-to-English dataset:** 979k parallel utterance pairs<br>**Fisher Spanish-to-English dataset: Spanish-English** | **Conversational Spanish-to-English dataset:**<br><br>**BLEU scores:**<br>**Source + Target model:** 42.7,<br>Cascaded model (STT & TTS): 48.7,<br>**Ground truth:** 74.7<br><br>**Fisher Spanish-to-English** | Mel-spectrogram (MFCCs) | **Advantages**<br><br>1. It explore different model trainings.<br>2. It utilises multi-head attention for speech-to-speech translation.<br><br>**Limitation**<br><br>1. The computational time of the algorithm was not considered. |

## 3.3 Summary of Literature Review

**Table 1:** Review of Related Works (Cont.)

| Ref | METHODS | DATASETS/LANGUAGE | RESULTS | FEAATURE EXTRACTION METHODS | ADVANTAGES / LIMITATIONS |
|---|---|---|---|---|---|
| (Jia et al., 2022) | Direct speech-to-speech<br><br>Translatotron 2 | Datasets:<br>Conversational es→en,<br>Fisher es→ *en*,<br>Multilingual CoVoST2 (es, fr, de, ca→en)<br>**Languages:**<br>Spanish-English, French-English, German-English, Catalan-English | Translation Quality<br>Fisher dataset:<br>outperformed Translatotron by 15.5 BLEU score.<br>Subjective MOS Naturalness Fisher dataset:<br>Translatotron: $3.70 \pm 0.08$<br>Translatotron2: $3.98 \pm 0.08$ | Mel-spectrogram features | Problems it addresses from the original Translatotron:<br>1. Non-contribution of the attention alignment learned. Advantages<br>1. Performance was closer to the cascaded system, Limitations<br>1. it does not consider the discrete rep. |

# 3. LITERATURE REVIEW (CONT.)

## 3.3 Summary of Literature Review

**Table 1:** Review of Related Works (Cont.)

| Ref | METHODS | DATASETS/LANGUAGE | RESULTS | FEAATURE EXTRACTION METHODS | ADVANTAGES / LIMITATIONS |
|---|---|---|---|---|---|
| (Julius et. al., 2025) | Direct speech-to-speech (Transformer architecture)<br><br>Augmented Attention Mechanism (73 hours / 71 hours) | **Datasets:**<br><br>200, 023 audio recordings<br><br>**Source:** Primary Public domain<br><br>**Language:**<br><br>English-Yoruba | **BLEU SCORE:** 0.7,<br><br>**MOS:** 2.8,<br><br>**Translation Accuracy:** 98<br><br>**Original Transformer translation Accuracy:** 88.16% | **Spectrogram based model** | **Advantages:**<br>1. Augmented attention mechanism to capture speech representation such as the diacritics, prosody, intonation etc.<br>2. More attention head (42) utilized for training.<br>**Limitations:**<br>1. The dataset could be scaled up for enhanced performance.<br>2. training input data is spectrogram based.<br>3. A trade-off between choice of attention-head and complexity. |

# 3. LITERATURE REVIEW (CONT.)

## 3.3 Summary of Literature Review

**Table 1:** Review of Related Works (Cont.)

| Ref | METHODS | DATASETS/LANGUAGE | RESULTS | FEAATURE EXTRACTION METHODS | ADVANTAGES / LIMITATIONS |
| --- | --- | --- | --- | --- | --- |
| (Prabhjot et. Al., 2024) | Sequence-to-sequence model: Direct Unit-to-Unit Translation (U2UT) model | **Datasets:**<br>1. FLEURS with 1625 parallel Punjabi-English speech.<br>2. Synthetically generated English speech<br>**Language:**<br>Punjabi-English | **U2UT Model:** **BLEU score:** 3.9829<br>**S2UT Model:** **BLEU score:** 0.2954<br>**Cascaded Model (Whisper-large-v3):** **BLEU score:** 18.1136 | **U2UT Model:**<br>No feature extraction was utilised.<br>HuBERT Base model trained on 960 hours of Librispeech | **Advantages:**<br>1. The use of discrete speech units that reduces the dimension of the speech signal while also preserving the original<br>**Limitations:**<br>1. The size of the dataset was not sufficient enough for training; hence performance of the proposed approach was lower than the cascaded system implemented with Whisper-large-v3. |

# 3. LITERATURE REVIEW (CONT.)

## 3.3 Summary of Literature Review

**Table 1:** Review of Related Works (Cont.)

| Ref | METHODS | DATASETS/LANGUAGE | RESULTS | FEAATURE EXTRACTION METHODS | ADVANTAGES / LIMITATIONS |
|---|---|---|---|---|---|
| (Radford et al., 2023) | Transformer encoder & decoder using MLP<br><br>AdamW optimiser | LibriSpeech, TED-LIUM 3, Common Voice 5.1, Artie bias Corpus, Call Home & Switch board, WSJ, CORAAL, CHiME-6, AMI-IHM AMI-SDM1, Meanwhile, Rev16, Kincald 46 Earnings-21,etc.<br>**LANGUAGE:** Non specific, multilingual ASR | **Transcription Multilingual LibriSpeech:**<br><br>**Whisper large-v2** (Spanish): WER = 4.2 %,<br>**Common Voice 9: Whisper large-v2** Dutch: WER = 5.8 %<br>Spanish: WER = 5.6 %<br>**FLEURS:** | GELU activation function,<br><br>MFCC | **ADVANTAGES**<br><br>1. Robust to noise (additive noise & pub noise)<br><br>**LIMITATIONS**<br><br>1. Ideal robustness could not be achieved.<br><br>2. Performance varied across different dataset. |

# 3. LITERATURE REVIEW (CONT.)

## 3.3 Summary of Literature Review

**Table 1:** Review of Related Works (Cont.)

| Ref | METHODS | DATASETS/LANGUAGE | RESULTS | FEAATURE EXTRACTION METHODS | ADVANTAGES / LIMITATIONS |
|---|---|---|---|---|---|
| **(Rubenstein et. al., 2023)** | Discretisation + Decoder-only transformer model<br>Models:<br>AudioPaLM 8B AST,<br>AudioPaLM 8B S2ST,<br>AudioPaLM-2 8B AST, | Datasets:<br>CoVoST2, CVSS, VoxPopuli, Common Voice, etc.<br>Language:<br>Nonspecific | CoVost2 Dataset (AST Task):<br>BLEU scores:<br>, AudioPaLM-2 8B AST: 37.8,<br>AudioPaLM-2 8B cascaded ASR + translation: 39.0<br>Whisper Large-v2: 29.1 | No feature extraction<br><br>Discrete tokenisation | **Advantages:**<br>1. It performed better than the SOTA Whisper-Large-v2 Model.<br>**Limitations:**<br>1. It is not a sequence-to-sequence modelling,<br>2. There might be possible delays introduced during training. |

## 3.3 Summary of Literature Review

**Table 1:** Review of Related Works (Cont.)

| Ref | METHODS | DATASETS/LANGUAGE | RESULTS | FEAATURE EXTRACTION METHODS | ADVANTAGES / LIMITATIONS |
|---|---|---|---|---|---|
| (Olatunji et al., 2023) | Collection of texts / transcripts from sites such as PubMed, NCBI, wiki-texts, development of audio recording web page. | Datasets: AfriSpeech<br>Language: English language of African accent from 13 countries. | AfriSpeech-clinical data<br>Whisper medium: WER of 0.568<br>Wave2Vec large: WER of 0.415<br>LibriSpeech(test-clean)<br>Whisper large pre-trained model: WER of 0.167 | Nill | **Advantages:**<br>1. Accented corpus.<br>2. Curated for general and medical domain.<br>**Limitations:**<br>1. Not curated for African indigeneous languages. |

# 3. LITERATURE REVIEW (CONT.)

## 3.3 Summary of Literature Review

**Table 1:** Review of Related Works (Cont.)

| Ref | METHODS | DATASETS/LANGUAGE | RESULTS | FEAATURE EXTRACTION METHODS | ADVANTAGES / LIMITATIONS |
|---|---|---|---|---|---|
| **(Anmol Gulati et al., 2020)** | Conformer model for ASR. | Datasets: LibriSpeech<br>Language: English language. | Big Conformer model with 118 million model parameters without a LM: WER = 0.021<br>With LM: WER = 0.019 | Spectrogram-based | **Advantages:**<br>1. Ability to capture long term dependencies (global) by transformer and local relationship by th help of CNN<br>**Limitations**<br>1. Just LSTM decoder was utilized for the end-to-end ASR model developed. |

# 3. LITERATURE REVIEW (CONT.)

## 3.4 Gaps in the Literature

**Table 2:** Summary of Gaps in the Literature

| Consideration | Authors | Gaps |
|---|---|---|
| **Dataset** | (Bansal, 2019); (Kaur et al., 2024); (Cherednichenko, 2020); (Grabar & Cardon, 2019); (Qian et al., 2023); (Kaur et al., 2024) | Insufficient amount of medical and nonmedical parallel speech corpus for the modelling of speech-to-speech translation<br>Difficulty in acquiring parallel speech corpus |
| **Complexity of the architectures** | (Jia et al., 2022) | The system is complex due to the architecture implemented, and the computational complexity of the models is not mostly stated |

# 3. LITERATURE REVIEW (CONT.)

## 3.4 Gaps in Literature

**Table 2:** Summary of Gaps in Literature (Cont.)

| Consideration | Authors | Gaps |
|---|---|---|
| **Speech discretisation** | (Chang et al., 2023)<br>(Rubenstein, Asawaroengchai, Nguyen, Bapna, Borsos, Quitry, Chen, Badawy, Han, Kharitonov, & others, 2023). | Limitation on work done<br>Possible generation of errors due to lack of discretisation |
| **Cross-lingual voice transfer** | (Jia, Weiss, et al., 2019);<br>(Kaur et al., 2024) | Challenges are encountered with cross-language voice transfer, and performance was lesser than the cascaded system |

# 3. LITERATURE REVIEW (CONT.)

## 3.4 Gaps in Literature

**Table 2:** Summary of Gaps in Literature (Cont.)

| Consideration | Authors | Gaps |
|---|---|---|
| **Complexity of the architectures** | (Jia et al., 2022) | The system is complex due to the architecture implemented, and the computational complexity of the models is not mostly stated |

# 4. METHODOLOGY

**Table 3: Table of Methodology to Achieve Specific Objectives**

| Specific Objectives | Methodology | Data | Expected Results |
|---|---|---|---|
| 1 | State-of-the-art text-to-text translation tools, Recording of the parallel corporal of both languages using a user friendly web site, Use of SPSS/MATLAB for analysis. | English text and corresponding Yorùbá text, Crowdsourced Human beings. | Quality accented English-Yorùbá parallel corpus |
| 2 | Sampling, Feature extraction: MFCCs, Discretisation: Hubert Modelling, K-means clustering, | Outputs from Objective 1 | Discretised features with reduced dimension ready to be fed as input to the model. |

# 4. METHODOLOGY

**Table 3: Table of Methodology to Achieve Specific Objectives**

| Specific Objectives | Methodology | Data | Expected Results |
|---|---|---|---|
| 3 | Sequence-to-modelling using transformer architecture, and training | Outputs from Objective 2 | State-of-the-art English-Yorùbá language translation model with |
| 4 | Utilisation of various software packages like the Python, HTML,CSS, APIs etc. | Output from Objective 3 | User friendly Softbots language translator. |
| 5 | Computation of BLEU scores, and MOS Naturalness and MOS Similarity, as well as testing of Softbot by selected individual | Outputs from Objective 3 and 4 | SOTA BLEU scores and MOS that is typical of a S2ST models. |

# 4. METHODOLOGY

## 4.1 Proposed Architecture

- **Figure 8 is the architecture of this proposed study to provide a broader understanding of the work.**

- **An improvement on Translatotron 2 (Figure 6)**

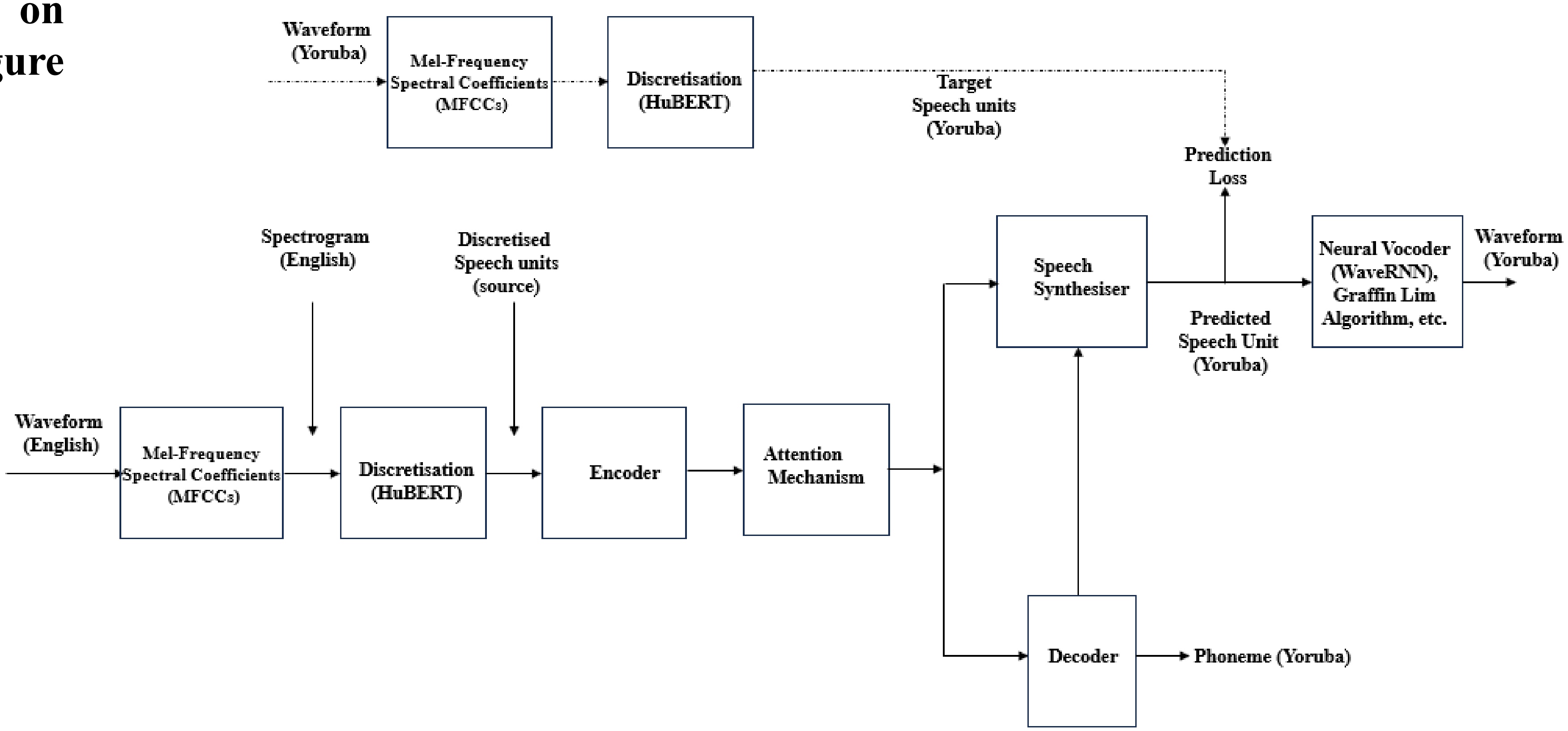



**Figure 8:**The Proposed Speech-To-Speech Neural Machine Translation Architecture

# 4. METHODOLOGY

## 4.1 Proposed Architecture (Cont.)

- The speech synthesizer of Figure 8 is shown in Figure 9.
- It is an extension to Translatotron 2, a SOTA speech-to-speech model.

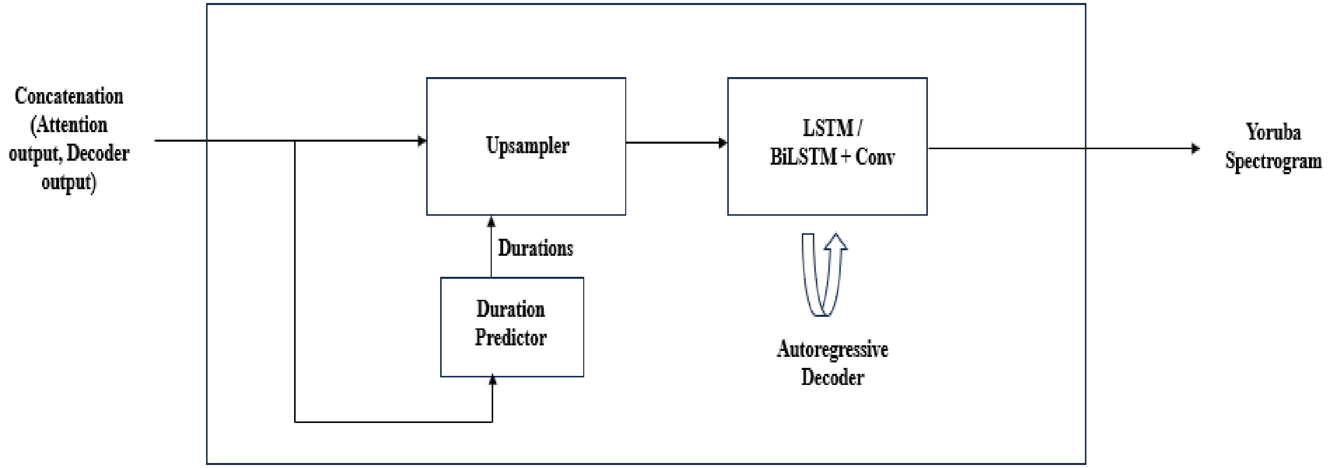



**Figure 9:** The Speech Synthesizer of Figure 8

### 3.1.1 Major Extension to be carried out

- English -Yoruba language pair corporal to be utilised for training.
- Speech discretization to be utilised unlike MFCC features, for both source and target languages.
- Stacks of LSTM, BiLSTMs shall be consider for the encoder and decoder.
- LSTM, BiLSTM shall be considered for the decoder of the speech synthesizer along with convolutional network.

# 4. METHODOLOGY

## 4.1 Proposed Architecture (Cont.)

Table 4: Translatotron2 versus Proposed Architecture

| S/N | Varying Blocks | Translatotron2 | Proposed Architecture |
|---|---|---|---|
| 1 | Training data nature | Spectrogram based | discretised |
| 2 | Encoder | Stacks of Conformer blocks:<br>-Feed-forward layer<br>-self-attention layer<br>-convolutional layer<br>-second feed-forward layer | -Same encoder as used in a transformer architecture (stack) (Vaswani et al., 2017), also to experiment with conformer block<br><br>-Stack of LSTM |
| 3 | Linguistic Decoder | Stack of LSTM | Stack of LSTM/BLSTM/GRU |
| 4 | Synthesizer | Stack of LSTM | Stack of LSTM/BLSTM/GRU |
| 5 | Source / Target language | Spanish / English | English / Yoruba |

# 4. METHODOLOGY

## 4.1 Proposed Architecture (Cont.)

### 4.1.1 Encoder

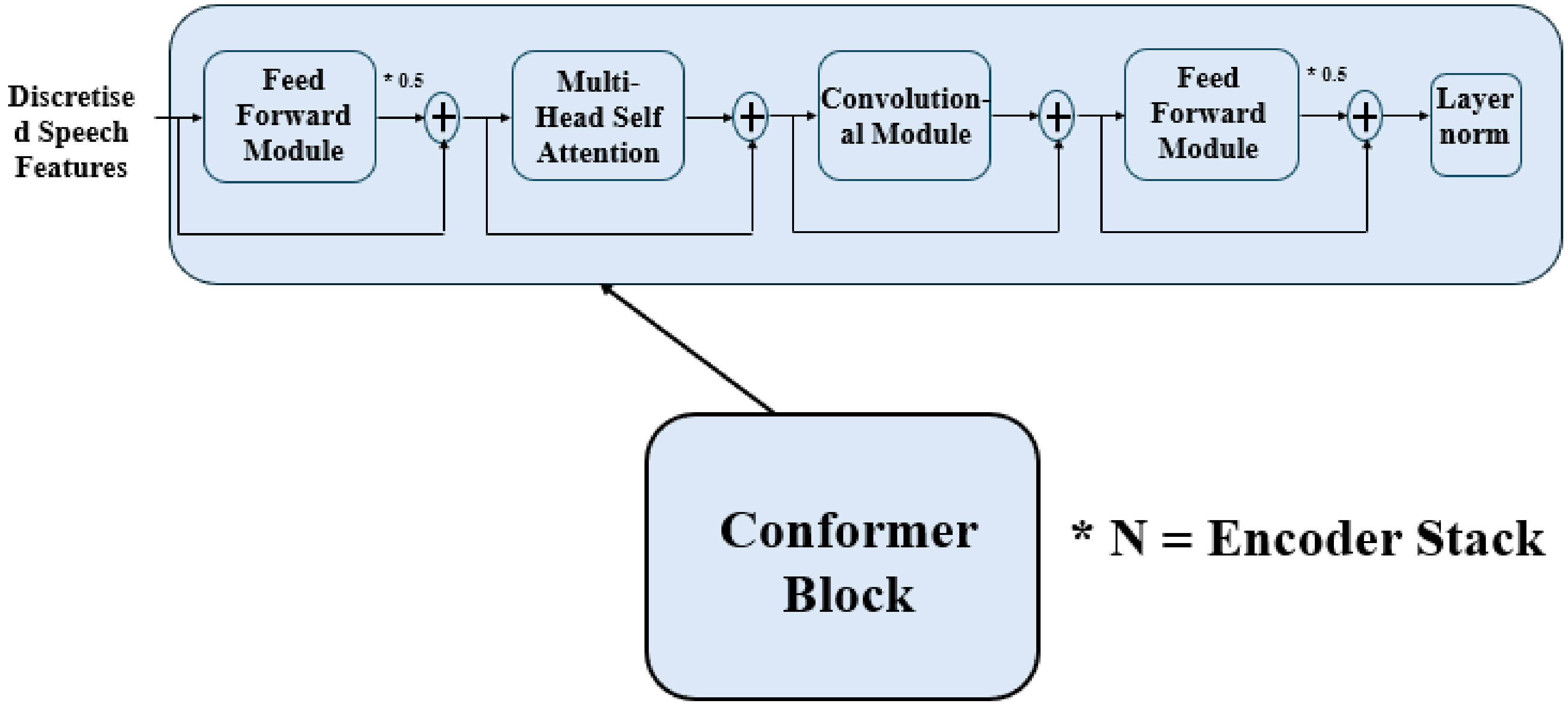



**Figure 10:** Conformer Block

# 4. METHODOLOGY

## 4.1 Proposed Architecture (Cont.)

### 4.1.1 Encoder

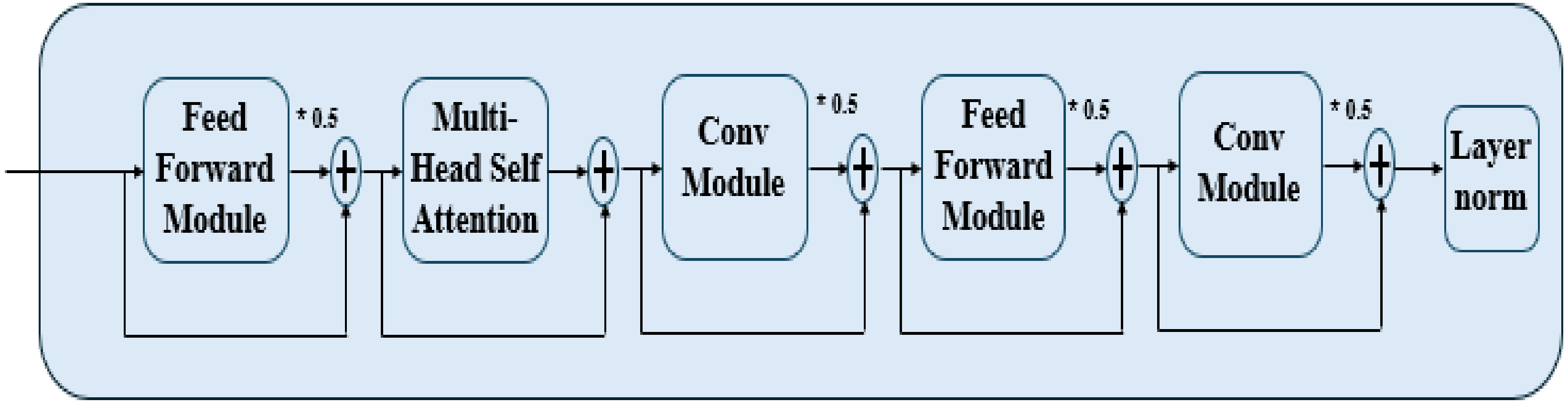



**Figure 11:** Conformer Block with Altered Convolutional Module

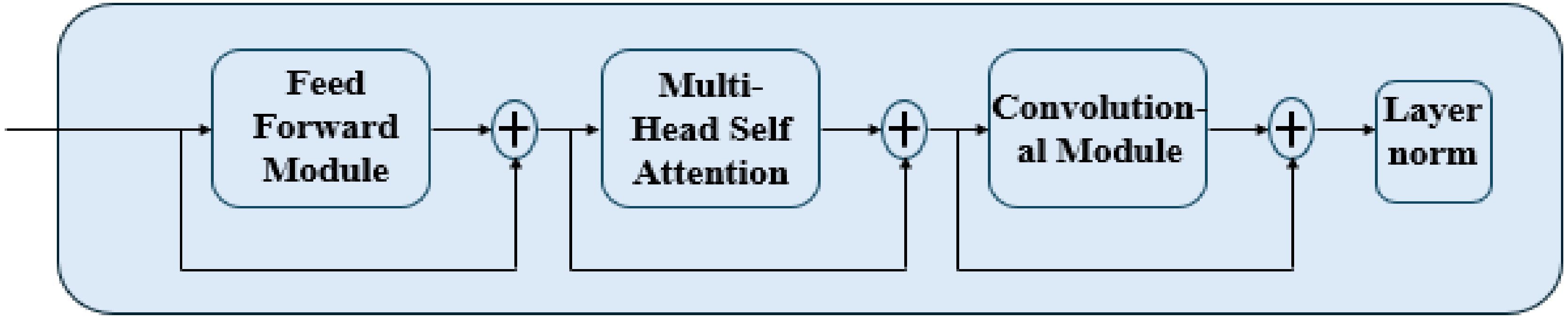



**Figure 12:** Conformer Block with Altered Feed Forward Module

# 4. METHODOLOGY

## 4.1 Proposed Architecture (Cont.)

### 4.1.1 Encoder

A. Feed Forward Module

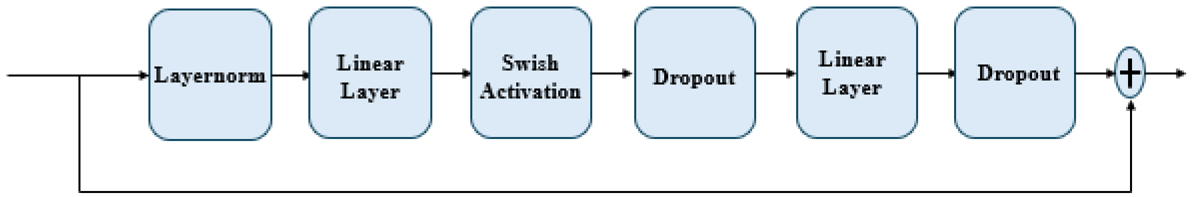



**Figure 13:** Feed Forward Module

C. Convolutional Module

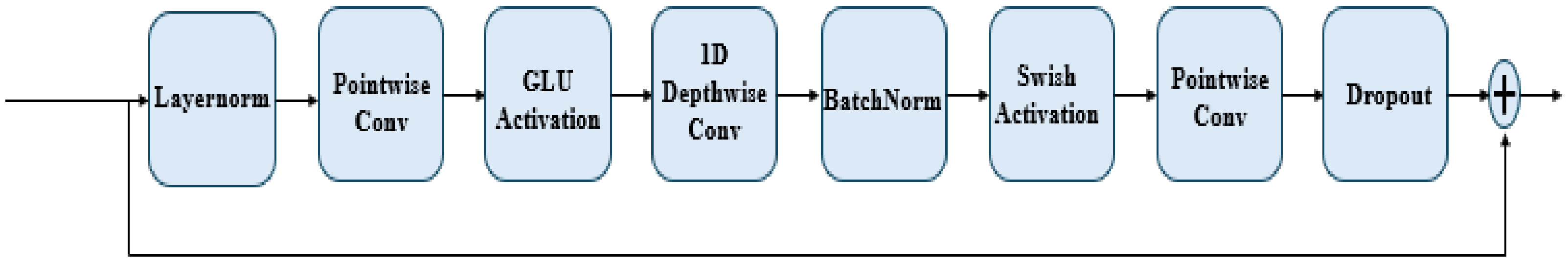



**Figure 14:** Convolutional Module

B. Multi-Head Self Attention (MHSA) Module

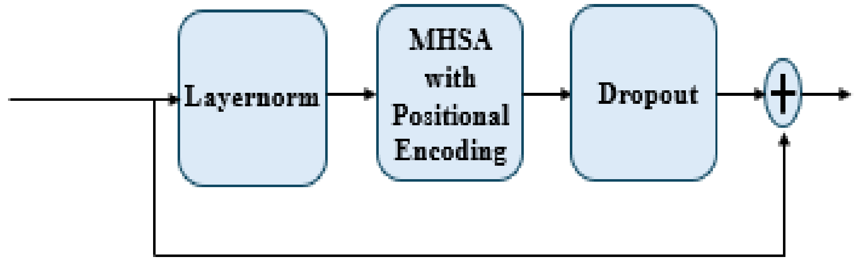



**Figure 15:** Multi-Head Self Attention Module

# 4. METHODOLOGY

## 4.1 Proposed Architecture (Cont.)

### 4.1.1 Encoder

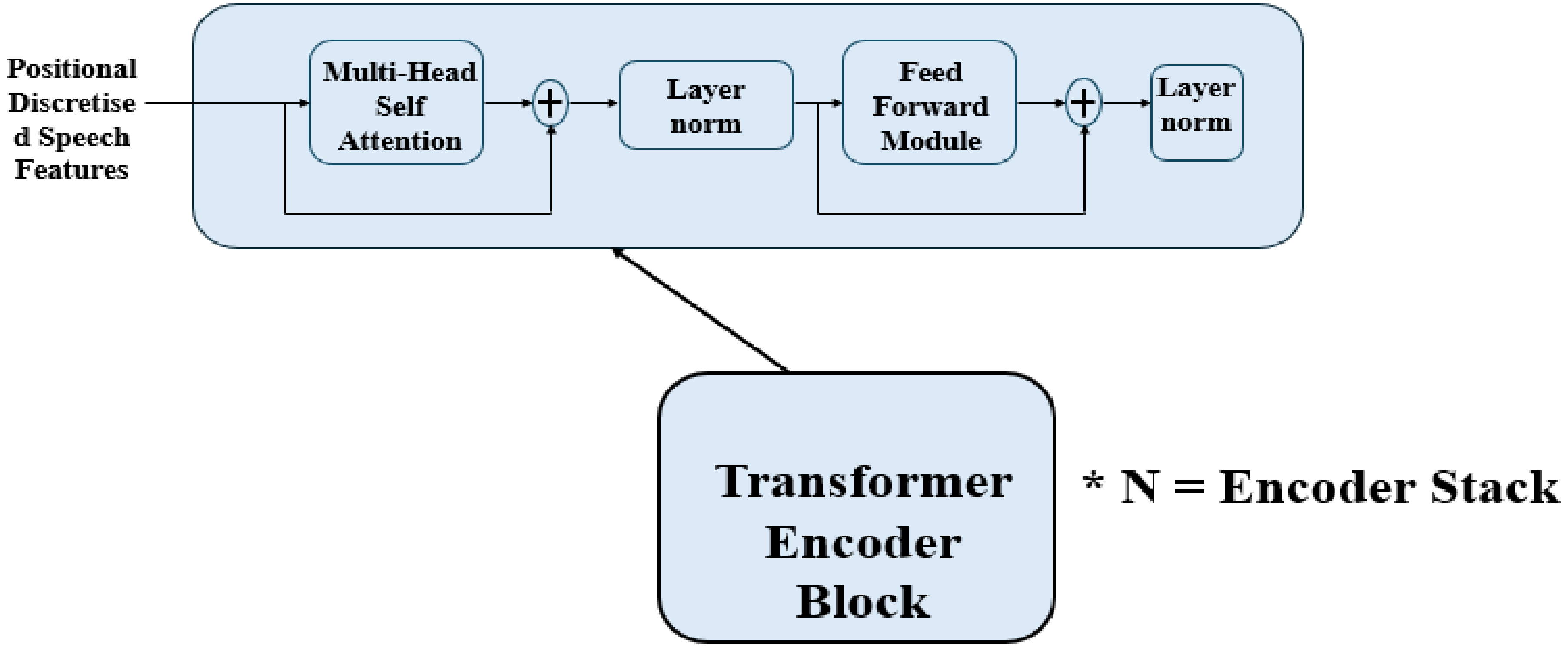
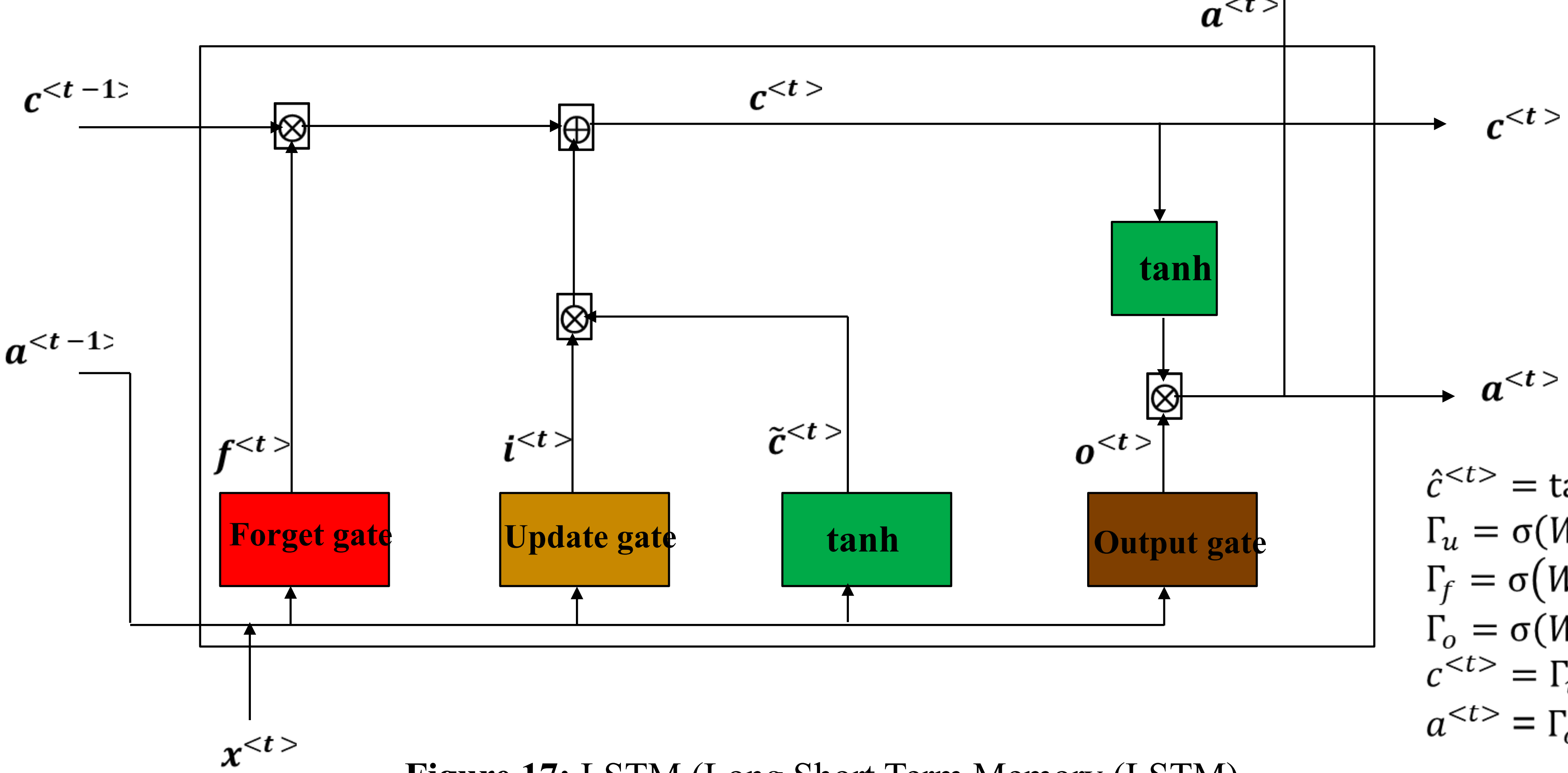



**Figure 16:** Transformer Encoder

# 4. METHODOLOGY

## 4.1 Proposed Architecture (Cont.)

### 4.1.2 Decoder (Stack of LSTMs)



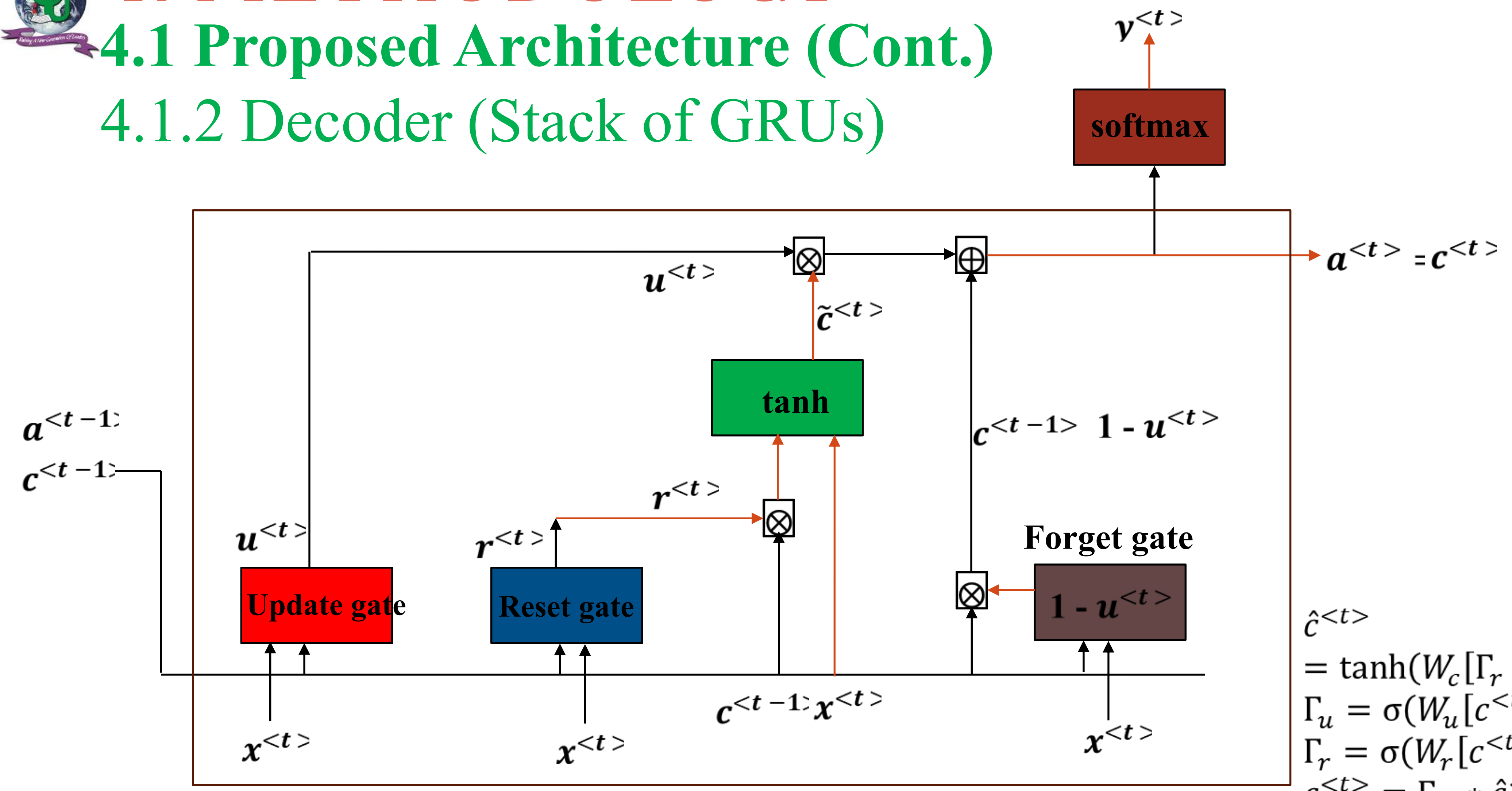

$$\hat{c}^{<t>} = \tanh(W_c[a^{<t-1>}, x^{<t>}] + b_c)$$
$$\Gamma_u = \sigma(W_u[a^{<t-1>}, x^{<t>}] + b_u)$$
$$\Gamma_f = \sigma\left(W_f[a^{<t-1>}, x^{<t>}] + b_f\right)$$
$$\Gamma_o = \sigma(W_o[a^{<t-1>}, x^{<t>}] + b_o)$$
$$c^{<t>} = \Gamma_u * \hat{c}^{<t>} + \Gamma_f * c^{<t-1>}$$
$$a^{<t>} = \Gamma_o * \tanh c^{<t>}$$

**Figure 17:** LSTM (Long Short Term Memory (LSTM)

# 4. METHODOLOGY

## 4.1 Proposed Architecture (Cont.)

### 4.1.2 Decoder (Stack of GRUs)




$$\hat{c}^{<t>} = \tanh(W_c[\Gamma_r * c^{<t-1>}, x^{<t>}] + b_c)$$

$$\Gamma_u = \sigma(W_u[c^{<t-1>}, x^{<t>}] + b_u)$$

$$\Gamma_r = \sigma(W_r[c^{<t-1>}, x^{<t>}] + b_r)$$

$$c^{<t>} = \Gamma_u * \hat{c}^{<t>} + (1 - \Gamma_u) * c^{<t-1>}$$

$$a^{<t>} = c^{<t>}$$

**Figure 18:** LSTM (Gated Recurrent Unit (GRU)

# 4. METHODOLOGY

## 4.1 Proposed Architecture (Cont.)

### 4.1.2 Decoder (Transformer Decoder)

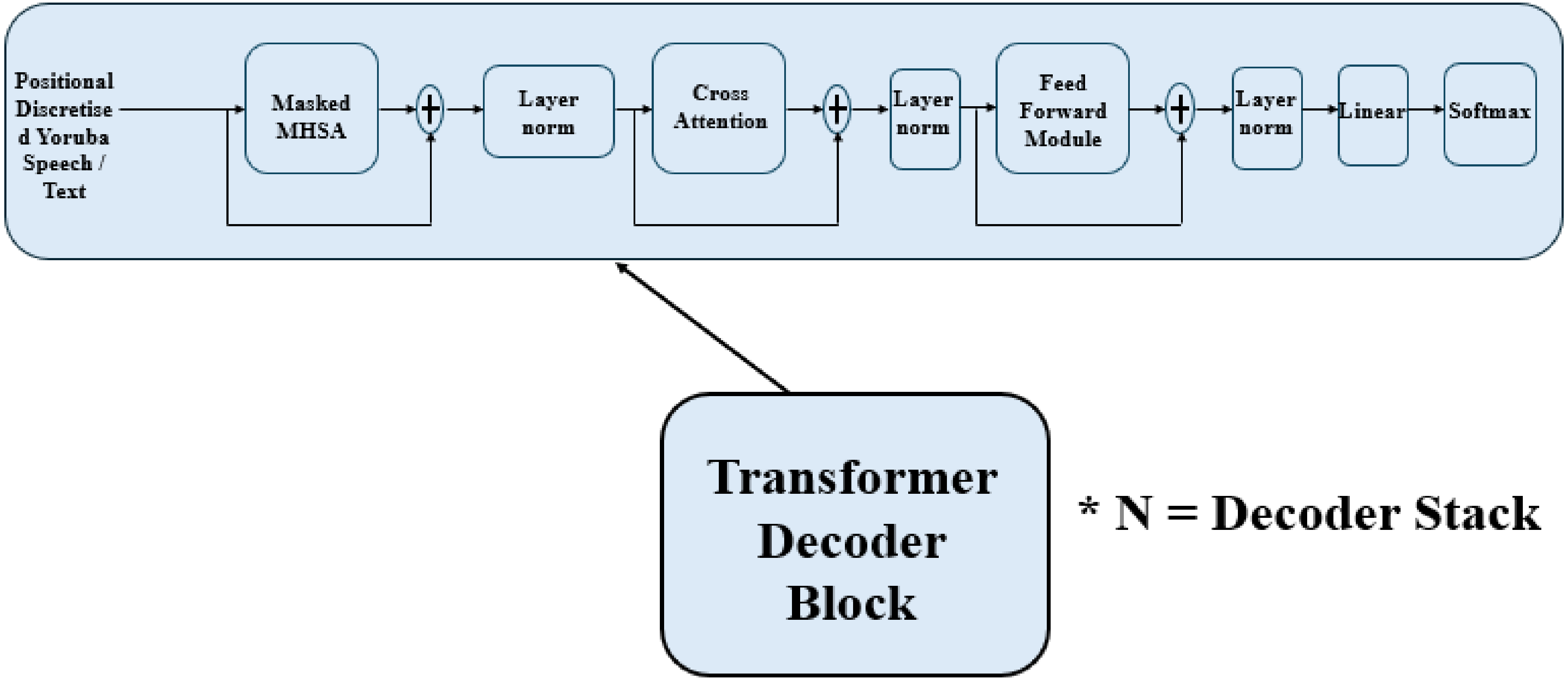



**Figure 19: Transformer Decoder**

# 4. METHODOLOGY

## 4.2 Flow Chart of the Proposed Model

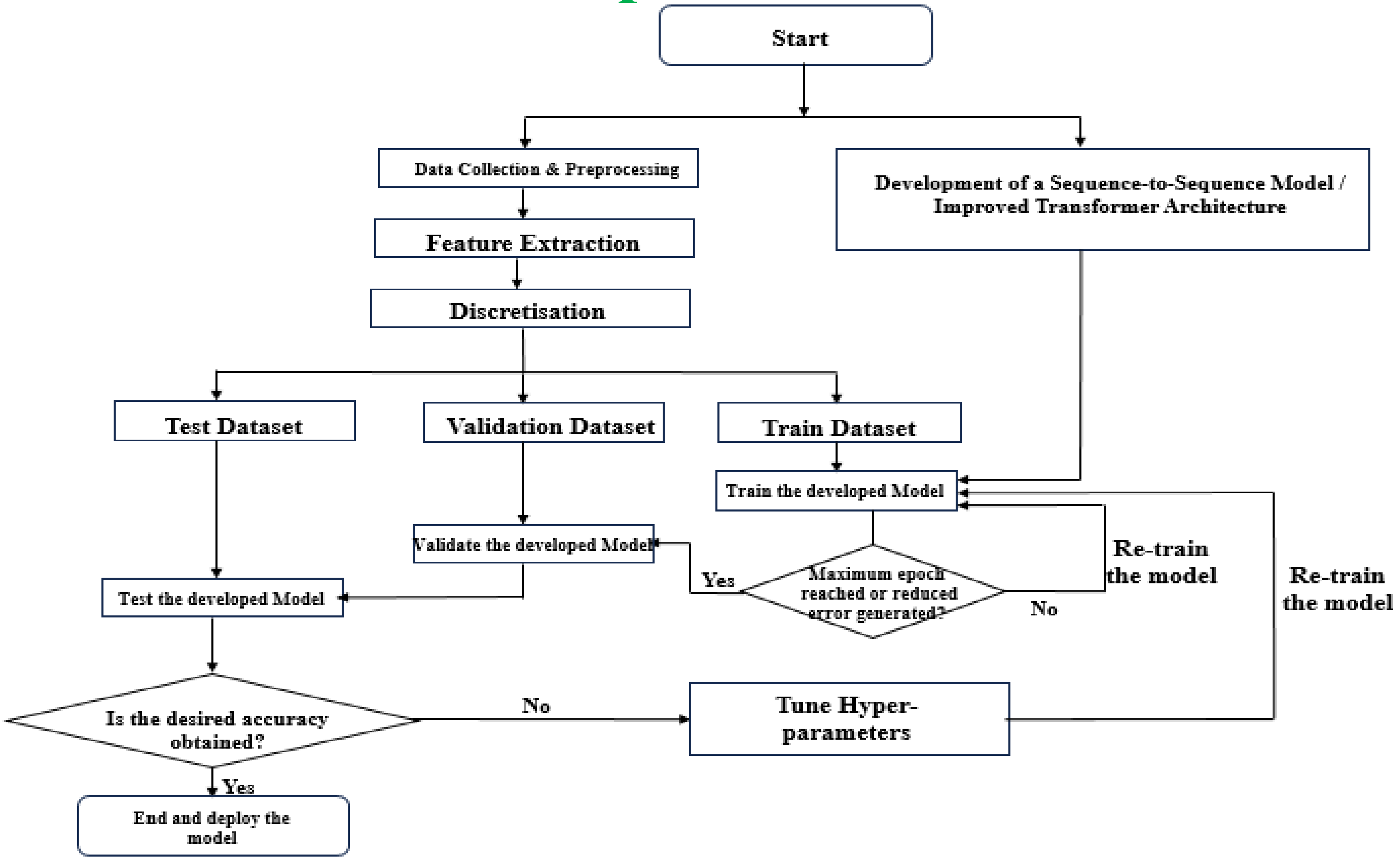



**Figure 20:** The Flowchart of the proposed Model

# 4.3 Outcomes of Method 1: Data Curation Feedback
## 4.3.1 Overall Statistics (03/03/2025 – 10/04/2026)

**Table 5: Overall Statistics [200,350 audios (444.10 hrs.); Speakers (505 + 8); # 5,264,124.9 – 10/04/2026]**

| S/N | Institutions | No of participants Projected | No of Registered participants | No of participants that completed | Amount Spent |
|---|---|---|---|---|---|
| 1 | UNILAG | 100 | Whatsapp – 50; Online - 38 | 22 (PH: 11; GEN: 11) + 5 | # 434,124.90 |
| 2 | UI | 100 | Whatsapp – 159; Online – 143, New Whatsapp – 194 | 84 (PH: 50; GEN: 34) + 2 | # 860,000.00 |
| 3 | FUNAAB | 100 | Whatsapp – (99+40); Online – 59 | 45 (PH: 20; GEN: 25) | # 450,000.00 |
| 4 | LASU | 100 | Whatsapp – 90; Online - 88 | 5 (PH: 1; GEN: 4) | # 50,000.00 |
| 5 | OAU | 100 | Whatsapp – None; Online – None | None | None |
| 6 | UI_2 + UI_3 | 100 | Whatsapp – 200+; Online – not yet verified | 96 (PH: 57; GEN: 39)<br>99 (PH: 96; GEN: 3) | # 960,000.00<br># 990,000.00 |
| 7 | UI_BATCH2 | 100 | Whatsapp – 180; | 106 (PH: 45; GEN: 61) | # 1,060,000.00 |
| 8 | UI_BATCH2_2 | 100 | Whatsapp – 180+; | 46 (PH: 44; GEN: 02) | # 460,000.00 |

# 4.3 Outcomes of Method 1: Data Curation Feedback

## 4.3.2 University of Lagos

### Table 6: UNILAG Statistics (22 + 5 Speakers)

| | | |
|---|---|---|
| 1 | num_audios_src | 4253 |
| 2 | num_audios_tgt | 4253 |
| 3 | total_num_audios (src+tgt) | 8506 |
| 4 | len_audio(time)_src (secs) | 28961.746 (8 hrs.) |
| 5 | len_audio(time)_tgt (secs) | 29151.519 (8 hrs.) |
| 6 | len_audio(time)_total (src + tgt) (secs) | 58113.26 (16 hrs.) |
| 7 | num_words_src | 40404 |
| 8 | num_unique_words_src* | 16825 |
| 9 | num_words_tgt | 44247 |
| 10 | num_unique_words_tgt* | 16098 |
| 11 | num_of_characters_src | 193862 |
| 12 | num_of_characters_tgt | 171099 |
| 13 | num_lines_of_text_src | 4253 |
| 14 | num_lines_of_text_tgt | 4253 |
| 15 | num_lines_of_text_total (src + tgt) | 8506 |

# 4.3 Outcomes of Method 1: Data Curation Feedback

## 4.3.3 University of Ibadan, UI

**Table 7: UI Statistics (84 + 2 Speakers)**

| | | |
|---|---|---|
| 1 | num_audios_src | 16,545 |
| 2 | num_audios_tgt | 16,545 |
| 3 | total_num_audios (src+tgt) | 33090 |
| 4 | len_audio(time)_src (secs) | 114240.87 (31.73 hrs.) |
| 5 | len_audio(time)_tgt (secs) | 127335.19 (35.37 hrs.) |
| 6 | len_audio(time)_total (src + tgt) (secs) | 241576.08 (67.10 hrs.) |
| 7 | num_words_src | 164230 |
| 8 | num_unique_words_src* | 64501 |
| 9 | num_words_tgt | 178084 |
| 10 | num_unique_words_tgt* | 64869 |
| 11 | num_of_characters_src | 781462 |
| 12 | num_of_characters_tgt | 686416 |
| 13 | num_lines_of_text_src | 16,545 |
| 14 | num_lines_of_text_tgt | 16,545 |
| 15 | num_lines_of_text_total (src + tgt) | 33090 |

# 4.3 Outcomes of Method 1: Data Curation Feedback

## 4.3.3 University of Ibadan, UI (Cont.)

**Table 8: UI 2 Statistics (96 Speakers)**

| | | |
|---|---|---|
| **1** | **num_audios_src** | 19,089 |
| **2** | **num_audios_tgt** | 19,089 |
| **3** | **total_num_audios (src+tgt)** | 38,178 |
| **4** | **len_audio(time)_src (secs)** | 112,449.6225 (31.24 hrs.) |
| **5** | **len_audio(time)_tgt (secs)** | 129,585.15547 (36.00 hrs.) |
| **6** | **len_audio(time)_total (src + tgt) (secs)** | 242,034.77797 (67.24 hrs.) |
| **7** | **num_words_src** | 160,992 |
| **8** | **num_unique_words_src*** | 54,432 |
| **9** | **num_words_tgt** | 178,327 |
| **10** | **num_unique_words_tgt*** | 47,971 |
| **11** | **num_of_characters_src** | 899,072 |
| **12** | **num_of_characters_tgt** | 785,255 |
| **13** | **num_lines_of_text_src** | 19,089 |
| **14** | **num_lines_of_text_tgt** | 19,089 |
| **15** | **num_lines_of_text_total (src + tgt)** | 38,178 |

# 4.3 Outcomes of Method 1: Data Curation Feedback

## 4.3.3 University of Ibadan, UI (Cont.)

**Table 9: UI 3 Statistics (99 Speakers)**

| | | |
|---|---|---|
| 1 | num_audios_src | 19,670 |
| 2 | num_audios_tgt | 19,670 |
| 3 | total_num_audios (src+tgt) | 39,340 |
| 4 | len_audio(time)_src (secs) | 143,350.773567 secs (39.82 hrs.) |
| 5 | len_audio(time)_tgt (secs) | 171,491.200867 secs (47.64 hrs.) |
| 6 | len_audio(time)_total (src + tgt) (secs) | 314,841.958434 secs (87.46 hrs.) |
| 7 | num_words_src | 223,063 |
| 8 | num_unique_words_src* | 70,973 |
| 9 | num_words_tgt | 2261,647 |
| 10 | num_unique_words_tgt* | 63,274 |
| 11 | num_of_characters_src | 1,293,412 |
| 12 | num_of_characters_tgt | 1,182,595 |
| 13 | num_lines_of_text_src | 19,670 |
| 14 | num_lines_of_text_tgt | 19,670 |
| 15 | num_lines_of_text_total (src + tgt) | 39,340 |

# 4.3 Outcomes of Method 1: Data Curation Feedback

## 4.3.3 University of Ibadan, UI (Cont.)

### Table 10: UI_BATCH2 Statistics (108 Speakers)

| | | |
|---|---|---|
| 1 | num_audios_src | 21,525 |
| 2 | num_audios_tgt | 21,525 |
| 3 | total_num_audios (src+tgt) | 43,050 |
| 4 | len_audio(time)_src (secs) | 175,867.93564 secs (48.85 hrs.) |
| 5 | len_audio(time)_tgt (secs) | 215,312.30856 secs (59.81 hrs.) |
| 6 | len_audio(time)_total (src + tgt) (secs) | 391,180.2442 secs (108.66 hrs.) |
| 7 | num_words_src | 311,376 |
| 8 | num_unique_words_src* | 116,233 |
| 9 | num_words_tgt | 362,813 |
| 10 | num_unique_words_tgt* | 120,394 |
| 11 | num_of_characters_src | 1,563,223 |
| 12 | num_of_characters_tgt | 1,502,789 |
| 13 | num_lines_of_text_src | 21,525 |
| 14 | num_lines_of_text_tgt | 21,525 |
| 15 | num_lines_of_text_total (src + tgt) | 43,050 |

# 4.3 Outcomes of Method 1: Data Curation Feedback
## 4.3.3 University of Ibadan, UI (Cont.)

**Table 11: UI_BATCH2_2 Statistics (46 Speakers)**

| | | |
|---|---|---|
| 1 | num_audios_src | 9,159 |
| 2 | num_audios_tgt | 9,159 |
| 3 | total_num_audios (src+tgt) | 18,318 |
| 4 | len_audio(time)_src (secs) | 74,328.87496 secs (20.65 hrs.) |
| 5 | len_audio(time)_tgt (secs) | 89,973.838363 secs (24.99 hrs.) |
| 6 | len_audio(time)_total (src + tgt) (secs) | 164,302.713323 secs (45.64 hrs.) |
| 7 | num_words_src | 104,591 |
| 8 | num_unique_words_src* | 41,089 |
| 9 | num_words_tgt | 127,973 |
| 10 | num_unique_words_tgt* | 40,194 |
| 11 | num_of_characters_src | 609,082 |
| 12 | num_of_characters_tgt | 593,118 |
| 13 | num_lines_of_text_src | 9,159 |
| 14 | num_lines_of_text_tgt | 9,159 |
| 15 | num_lines_of_text_total (src + tgt) | 18,318 |

# 4.3 Outcomes of Method 1: Data Curation Feedback
## 4.3.4 Federal University of Agriculture Abeokuta, FUNAAB

**Table 12: FUNAAB Statistics (45 Speakers)**

| | | |
|---|---|---|
| 1 | num_audios_src | 8,953 |
| 2 | num_audios_tgt | 8,953 |
| 3 | total_num_audios (src+tgt) | 17,906 |
| 4 | len_audio(time)_src (secs) | 78,831.28686 (21.90 hrs) |
| 5 | len_audio(time)_tgt (secs) | 89,138.5185 (24.76 hrs) |
| 6 | len_audio(time)_total (src + tgt) (secs) | 167,969.80536 (46.66 hrs) |
| 7 | num_words_src | 102,197 |
| 8 | num_unique_words_src* | 36,157 |
| 9 | num_words_tgt | 111,779 |
| 10 | num_unique_words_tgt* | 37,566 |
| 11 | num_of_characters_src | 500,223 |
| 12 | num_of_characters_tgt | 445,647 |
| 13 | num_lines_of_text_src | 8,953 |
| 14 | num_lines_of_text_tgt | 8,953 |
| 15 | num_lines_of_text_total (src + tgt) | 17,906 |

# 4.3 Outcomes of Method 1: Data Curation Feedback
## 4.3.5 Lagos State University, LASU

**Table 13: LASU Statistics (5 Speakers)**

| | | |
|---|---|---|
| 1 | num_audios_src | 981 |
| 2 | num_audios_tgt | 981 |
| 3 | total_num_audios (src+tgt) | 1,962 |
| 4 | len_audio(time)_src (secs) | 8,633.685334 (2.39 hrs.) |
| 5 | len_audio(time)_tgt (secs) | 10,137.173334 (2.81 hrs.) |
| 6 | len_audio(time)_total (src + tgt) (secs) | 18,770.858664 (5.21 hrs.) |
| 7 | num_words_src | 12,373 |
| 8 | num_unique_words_src* | 4,691 |
| 9 | num_words_tgt | 13,016 |
| 10 | num_unique_words_tgt* | 5,012 |
| 11 | num_of_characters_src | 56,754 |
| 12 | num_of_characters_tgt | 50,552 |
| 13 | num_lines_of_text_src | 981 |
| 14 | num_lines_of_text_tgt | 981 |
| 15 | num_lines_of_text_total (src + tgt) | 1,962 |

# 4.3 Outcomes of Method 1: Data Curation Feedback
# 4.3.6 Percentage Collected

**Table 14: PERCENTAGE OF NUMBER OF AUDIOS COLLECTED**

**NUMBER OF AUDIOS PROJECTED = 200,000; OBTAINED PER LANGUAGE: 100,175 AUDIOS**

| | UNILAG | UI | FUNAAB | LASU | UI_2 | UI_3 | UI_Batch2 |
|---|---|---|---|---|---|---|---|
| **Total number of EN** | 4,253 /18,181.81 (23.39%) | 16,545 /18,181.81 (91.00%) | 8,953/18,181.81 (49.24%) | 981/18,181.81 (5.40%) | 19,089/18,181.81 (104.99%) | 19,670/18,181.81 (108.19%) | 21,525/18,181.81 (118.39%) |
| **Total number of YOR** | 4,253 /18,181.81 (23.39%) | 16,545 /18,181.81 (91.00%) | 8,953/18,181.81 (49.24%) | 981/18,181.81 (5.40%) | 19,089/18,181.81 (104.99%) | 19,670/18,181.81 (108.19%) | 21,525/18,181.81 (118.39%) |
| **Overall %** | 4253 + 16545 + 8,953 + 981 + 19,089 + 19,670 + 21,525 + 9,159 = 100,175 / 200,000 = **50.09%** | | | | | | |

# 4.3 Outcomes of Method 1: Data Curation Feedback
## 4.3.6 Percentage Collected

**Table 14: PERCENTAGE OF NUMBER OF AUDIOS COLLECTED (CONT.)**

**NUMBER OF AUDIOS PROJECTED = 200,000; OBTAINED PER LANGUAGE: 100,175 AUDIOS**

| | UI_Batch2_2 | | | | | | |
|---|---|---|---|---|---|---|---|
| **Total number of EN** | 9,159/18,181.81 (50.37%) | | | | | | |
| **Total number of YOR** | 9,159/18,181.81 (50.37%) | | | | | | |
| **Overall %** | 4253 + 16545 + 8,953 + 981 + 19,089 + 19,670 + 21,525 + 9,159 = 100,175 / 200,000 = **50.09%** | | | | | | |

# 4.3.6 Percentage Collected (Contd)

**Table 15: PERCENTAGE OF HOURS OF AUDIOS COLLECTED**

**HOURS OF AUDIOS PROJECTED = 1,000 hrs.;**
**OBTAINED: 204.62 hrs. (EN), 239.48 hrs. (YOR), TOTAL = 444.10 hrs.**

| | UNILAG | UI | FUNAAB | LASU | UI_2 | UI_3 | UI_Batch 2 |
|---|---|---|---|---|---|---|---|
| **Total hours of EN** | 8.04 | 31.73 | 21.90 | 2.39 | 31.24 | 39.82 | 48.85 |
| **Total hours of YOR** | 8.10 | 35.37 | 24.76 | 2.81 | 36.00 | 47.64 | 59.81 |
| **Total hours (Expected = 90.90 hrs. (9.09%))** | 16.14 (1.61%) | 67.10 (6.71%) | 46.66 (4.67%) | 5.21 (0.52%) | 67.24 (6.72%) | 87.46 (8.75%) | 108.66 (10.86%) |
| **Overall %** | 16.14 + 67.10 + 46.66 + 5.21 + 67.24 + 87.46 + 108.66 + 45.64 = 444.10 / 1,000 = **44.41 %** | | | | | | |

# 4.3.6 Percentage Collected (Contd)

**Table 15: PERCENTAGE OF HOURS OF AUDIOS COLLECTED (CONT.)**

**HOURS OF AUDIOS PROJECTED = 1,000 hrs.;**
**OBTAINED: 204.62 hrs. (EN), 239.48 hrs. (YOR), TOTAL = 444.10 hrs.**

| | UI_Batch2_2 | | | | | | |
|---|---|---|---|---|---|---|---|
| **Total hours of EN** | 20.65 | | | | | | |
| **Total hours of YOR** | 24.99 | | | | | | |
| **Total hours (Expected = 90.90 hrs. (9.09%))** | 45.64 (4.56%) | | | | | | |
| **Overall %** | 16.14 + 67.10 + 46.66 + 5.21 + 67.24 + 87.46 + 108.66 + 45.64<br>= 444.10 / 1,000 = **44.41 %** | | | | | | |

**FROM 22 + 84 + 45 + 5 + 96 + 99 + 108 + 46 = 505 DOCUMENTED SPEAKERS / 1100 OBTAINABLE; % OF SPEAKERS = 45.90 %**

# 4. METHODOLOGY (CONT.)

## 4.4 Budget

**Table 16: Research Budget**

| S/n | ITEM | COST ($) | QUANTITY | TOTAL ($) |
|---|---|---|---|---|
| 1 | Speech Data Curation (Google PhD) | | 10 weeks | 10,064.14 |
| 3 | Conferences (CApIC ACE) | 750 | 2 | 1,500 |
| 4 | Publications (CApIC ACE) | 2000 | 2 | 4,000 |
| | Total | | | 15,564.14 |

# 4.5 Timelines

**Table 17: Research Timeline Detailing Semester Breakdown of Activities**

| TASK NO | TASK NAME | START DATE | END DATE | OMEGA SEMESTER, 2023/2024 | ALPHA SEMESTER 2024/2025 | OMEGA SEMESTER 2024/2025 | ALPHA SEMESTER 2025/2026 | OMEGA SEMESTER 2025/2026 |
|---|---|---|---|---|---|---|---|---|
| 1 | Development of Proposal | 04/01/2024 | 01/06/2024 | Yes | | | | |
| 2 | Field Work | 09/04/2025 | 30/03/2026 | | Yes | Yes | Yes | |
| 3 | Methodology | 09/04/2025 | 30/03/2026 | | | Yes | Yes | |
| 4 | Results and Discussion | 01/01/2026 | 30/03/2026 | | | | Yes | |
| 5 | Completion of Report | 01/04/2026 | 27/08/2026 | | | | | Yes |

## 4.5.1 Field Work

| 09/04/2025 | 30/03/2026 |
|---|---|

| OMEGA SEMESTER 2024/2025 | ALPHA SEMESTER 2025/2026 |
|---|---|

# Table 18: Field Work Breakdown

| APRIL-JUNE, 2025 | JULY-SEPTEMBER, 2025 | OCTOBER, 2025 | NOVEMBER-DECEMBER, 2025 | JANUARY, 2026 | FEBRUARY, 2026 | MARCH, 2026 |
|---|---|---|---|---|---|---|
| **Generation / Curation of dataset**<br><br>**Analysis of dataset** | **Generation / Curation of dataset**<br><br>**Understanding of datasets / installation of software packages, modules**<br><br>**Analysis of dataset** | **Generation / Curation of dataset**<br><br>**Analysis of dataset** | **Generation / Curation of dataset**<br><br>**Analysis of dataset** | **Analysis of dataset** | **Analysis of dataset** | **Performance analysis of the developed models: Base and Transfer Learning models** |
| Sending out of notification message to intended individual for the task | Familiarisation with the installed software, and collected parallel audio files, as well as dataset available<br><br>Knowledge of Python programming for deep learning<br><br>Creation of manifest files | Familiarisation with python modules for the aforementioned models<br><br>Knowledge of Python programming for deep learning<br><br>Creation of manifest files | Preparation of dataset for public access<br><br>Statistical analysis<br><br>Creation of manifest files | Preparation of dataset for public access<br><br>Statistical analysis<br><br>Creation of manifest files | Compare the transfer learning results obtained with that of the BASE MODEL developed.<br><br>Preparation of dataset for public access<br><br>Statistical analysis<br><br>Creation of manifest files | Editing and proof reading of the whole of chapter 3 for corrections, and update<br><br>Preparation of dataset for public access<br><br>Statistical analysis<br><br>Creation of manifest files |
| Development of web site for collection of data through crowdsourcing | Crowdsourcing and recording in UNILAG, LASU, FUNAAB, and UI | Crowdsourcing and recording in OAU, FUTA, University of Ado-Ekiti, and UNILORIN | Crowdsourcing and recording in University of Lome, Togo, and Univ. of Abomey Calavi, Benin Republic | Crowdsourcing and recording in CU | | |
| Sending out of links for the collection of audio data to intended individual. | Thorough analysis of the dataset for deep understanding, and interpretation using matlab, spss, possibly R | Knowledge of discritisation like the HuBERT, Wav2Vec models and how they are implemented | | | | . |

## 4.5.2 Methodology

| 09/04/2025 | 30/03/2026 | OMEGA SEMESTER 2024/2025 | ALPHA SEMESTER 2025/2026 |
|---|---|---|---|

# Table 19: Methodology Breakdown

| APRIL-JUNE, 2025 | JULY-SEPTEMBER, 2025 | OCTOBER, 2025 | NOVEMBER-DECEMBER, 2025 | JANUARY, 2026 | FEBRUARY, 2026 | MARCH, 2026 |
|---|---|---|---|---|---|---|
| **Generation / Curation of dataset**<br><br>**Development of website**<br><br>**Scraping online websites for texts transcripts**<br><br>**Use of text-to-text translation**<br><br>**Crowd-sourcing individuals for recording** | **Generation / Curation of dataset**<br><br>**Development of models for data preprocessing**<br><br>**Development of models for discretization, encoder, speaker encoder, attention mechanism, decoder, and vocoder** | **Generation / Curation of dataset**<br><br>**Development of models for data preprocessing**<br><br>**Development of models for discretization, encoder, speaker encoder, attention mechanism, decoder, and vocoder** | **Generation / Curation of dataset**<br><br>**Development of models for data preprocessing**<br><br>**Development of models for discretization, encoder, speaker encoder, attention mechanism, decoder, and vocoder** | **Integration of the developed models into a single base model**<br><br>**Train, Validate, Test, and carryout performance evaluation**<br><br>**Develop Transfer Learning model**<br><br>**Train, Validate, Test, and carryout performance evaluation** | **Train, Validate, Test, and carryout performance evaluation**<br><br>**Develop a prototype Softbot** | **Integration of developed models to the prototype softbot** |
| **Use of Python** Flask with HTML, CSS and Vercel server for recording webpage.<br><br>**Thorough search** for online media database, newspapers, etc.<br><br>**Use of Google** translate<br><br>**Crowdsource** individual from 11 institutions to participate in recordings | Same as previous<br><br>Analysis of data<br><br>Familiarizations with Python modules for the packages to implement the models.<br><br>Development of the models mostly taking into consideration the discretization with HuBERT, transformer models. | Same as January<br><br>Knowledge of the choice of the hyperparameters, for various models of the expected base model.<br><br>Gain further insight from literatures | Same as February<br><br>Test each of the developed models.<br><br>Make a choice of the language translation pretrained model to be utilised for transfer learning | **Split** the preprocessed discretised dataset into train, validate and test, **train** the training data using the single developed base, and transfer learning models **evaluate** using metrics like the MOS, BLEU, scores, precision, recall, F1-score,and ROUGE. **[using the FEDGEN platform to run the code]** | **Train** the training data using the single developed base and transfer learning model, **evaluate** using metrics like the MOS, BLEU, scores, precision, recall, F1-score,and ROUGE. **[using the FEDGEN platform to run the code] – For both medical and non medical corpus**<br><br>Ablation studies | **Development** of another website (Softbot) for interfacing with users.<br><br>**Use of APIs** to integrate the model output to the developed wed site interface.<br><br>**Sourcing** of raters to access the performance of the Softbot |

| 01/01/2026 | 30/03/2026 |
|---|---|

ALPHA SEMESTER 2025/2026

# Table 20: Results and Discussion Breakdown

| JANUARY, 2026 | JANUARY, 2026 | FRBRUARY, 2026 | FRBRUARY, 2026 | MARCH, 2026 | MARCH, 2026 |
|---|---|---|---|---|---|
| **Analysis of the results of the various experimental setup for transfer learning and base model for medical and non medical speech corpus** | **Analysis of the results of the various experimental setup for transfer learning and base model for medical and non medical speech corpus** | **Updating of the experimental parameters and extraction of new results** | **Development of write up for the discussion** | **Development of write up for the discussion** | **Thorough look through the developed chapter contents for experimental errors and possible corrections.** |
| I shall do both analytical illustration of the results.<br><br>And graphical illustration if needed. | I shall do both analytical illustration of the results.<br><br>And graphical illustration if needed. | Based on the already obtained results, there might need to update or enhance the performance of the models via change of hyperparameters of the model. | A coherent discussion of the obtained results will be carried out without bias. | A coherent discussion of the obtained results will be carried out without bias. | Editing and proof reading<br><br>Checking for experimental hyperparameters erroneously utlised for each of the experiments. |
| Use of MATLAB, Python for analysis, and interpretation of results | Use of MATLAB, Python for analysis, and interpretation of results | I shall adjust these parameters and report possible improvement if any or as a limitation of the study when certain parameters are adjusted. | Discussion of the findings (analysis, tables, possible chat) will be reported and the whole document of chapter 4 will be integrated as a single file. | Discussion of the findings (analysis, tables, possible chat) will be reported and the whole document of chapter 4 will be integrated as a single file. | Drawing the summary of the chapter. |

## Table 21: Completion of Report

| APRIL - JUNE, 2026 | JULY-AUGUST, 2026 |
|---|---|
| **Merging of all chapters into a single document** | **Submission of report** |
| Merge the proof read chapter 1, chapter 2, chapter 3, chapter 4, chapter 5, and appendix into a single document | Submission of the completed report for Viva |
| | Corrections of errors pointed out. |

## 4.5.5 Summary of Timeline

# Table 22: Summary of Research Timeline

| S/N | ACTIVITY | YEAR ONE | | | | | | | | | | | | YEAR TWO | | | | | | | | | | | | YEAR THREE | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 |
| 1 | Defining Research Focus | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 2 | Literature Review | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 3 | Developing Methodology | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 4 | Ethical Application | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 5 | Data Collection | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 6 | Data Analysis | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 7 | Proposal | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 8 | Thesis Writing | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 9 | Publication | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 10 | MILESTONE | | | | | Seminar1 Seminar2 | | | | | Departmental Proposal | | | College Proposal | | | | | Field Work | | | | | | | | | | | | Departmental and College Post Field Defence | | | Title Registration | Defence (Viva) | Correction and Submission of Corrected Thesis | |

# 5. CONCLUSION

- **The development of speech-to-speech modelling has seen revolution and improvement for high-resourced languages like English, Spanish, French, and German.**

- **However, the African indigenous languages, including the Yoruba language, are still underdeveloped in language translation from another language to Yoruba and vice versa.**

- **This proposed work investigates how to publicly make available parallel English-Yoruba speech corpora to aid further studies in model development for other researchers.**

- **This work will leverage the transfer learning approach to train medical speech corpus and non-medical speech corpus to obtain a robust speech translation output which will further guarantee the utilisation of speech translation in the medical space, as well as public engagement between medical health practitioners and public thereby addressing the SDGs 3 and 10 and the IR4.**

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